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(1) Research Program/Project/Study Title: Varietal Trial of the Different Sorghum (<i>Sorghum bicolor</i>) Organically Grown under Capiz Condition	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Sorghum	(8) Discipline/Field of Specialization Agriculture
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture Project of CapSU Burias, Mambusao, Capiz, Philippines. The experiment was laid out using the Completely Randomized Design with three replications. The treatments were the different varieties of sorghum which comprised of A – white varieties, B – red varieties, and C – purple varieties.

Growth and yield parameters such as plant height, stem girth, biomass, number of leaves, length of leaves, width of leaves, number of spikes per panicle, number of spikelets per spike, weight of panicle, length of panicle, width of panicle, number of grains per panicle, 1000 seeds weight, number of roots, weight of roots, and length of roots were gathered and analyzed using the Statistical Tool for Agricultural Research (STAR) software.

The different varieties significantly affected the parameters of sorghum except for the number of spikes per panicle, number of spikelets per spike, number of grains per panicle, number of roots, and length of roots. Purple varieties exhibited tallest plants, heaviest biomass, longest and widest leaves, heaviest 1000 seeds weight, and heaviest roots, widest panicle, longest panicle and heaviest panicles among all the varieties. Biggest stems and most number of leaves produced were observed from the red sorghum.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Sorghum, Capiz Condition

- 1. Introduction** (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

*Sorghum (*Sorghum bicolor*) is a versatile and resilient cereal crop known for its ability to thrive in diverse climatic conditions, making it a critical food and forage resource in many parts of the world, including the Philippines. The province of Capiz, situated in the Western Visayas region, presents unique agro-climatic conditions that can influence the performance of sorghum varieties. Conducting varietal trials is essential to identify the most suitable sorghum cultivars that can maximize yield and adapt to local conditions, thereby contributing to food security and agricultural sustainability.*

The adaptability and resilience of sorghum to various environmental stresses, such as drought and poor soil fertility, have been well documented (Miller, 1984; Rooney, 2000). This makes it an ideal crop for regions like Capiz, which may experience fluctuations in weather patterns and soil conditions. Sorghum's deep root system and efficient water use are key traits that enable it to withstand periods of drought, a common challenge in tropical and subtropical regions (Blum, 2004).

Varietal trials play a crucial role in evaluating the performance of different sorghum cultivars under specific environmental conditions. These trials help in assessing various agronomic traits, including yield, growth duration, resistance to pests and diseases, and tolerance to abiotic stresses (Kenga et al., 2004). In the context of Capiz, it is vital to consider local environmental factors such as soil type, rainfall patterns, and temperature variations when selecting sorghum varieties for trial (FAO, 1995).

Research has shown that the performance of sorghum varieties can vary significantly based on their genetic makeup and the environmental conditions in which they are grown. Studies by Tuinstra et al. (1997) and Reddy et al. (2006) highlight the importance of selecting sorghum varieties with specific traits tailored to local conditions. For instance, varieties with early maturity can be advantageous in regions with shorter growing seasons or unpredictable rainfall patterns.

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In the Philippines, previous studies have identified several sorghum varieties that show promise under local conditions. For example, the Bureau of Agricultural Research (BAR) has conducted trials on various sorghum cultivars, demonstrating the potential of specific varieties to perform well in different parts of the country, including Capiz (BAR, 2017). These studies emphasize the need for continuous evaluation and development of sorghum varieties to ensure they meet the evolving needs of local farmers and climatic conditions.

Moreover, the integration of sorghum into the cropping systems of Capiz can offer multiple benefits. Sorghum can serve as a supplementary crop, providing an alternative source of food, feed, and bioenergy, thus enhancing the overall resilience and productivity of the agricultural system (Smith & Frederiksen, 2000). The adoption of high-performing sorghum varieties can also contribute to crop diversification, reducing the risk of crop failure and improving food security in the region (Doggett, 1988).

In conclusion, conducting varietal trials of different sorghum cultivars under the conditions of Capiz, Philippines, is essential for identifying the most suitable varieties that can thrive in this specific agro-climatic environment. These trials not only help in optimizing yield and adaptability but also contribute to the sustainability and resilience of local agricultural systems. Continued research and collaboration among local institutions, farmers, and researchers are crucial for advancing the development and adoption of sorghum varieties that can meet the needs of the Capiz region.

Objectives of the Study

The study was conducted to determine performance of the different varieties of sorghum (*Sorghum bicolor*) under Capiz condition. It specifically aimed to:

1. Evaluate growth performance of the different sorghum;
2. Find out which sorghum variety will give the best result under the Capiz condition;
3. Recommend the best sorghum variety to plant under the Capiz condition.

(10) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted at the Organic Agriculture Project of CapSU Burias, Mambusao, Capiz, Philippines. The experiment was laid out using the Completely Randomized Design with three replications. The treatments were the different varieties of sorghum which comprised of A – white varieties, B – red varieties, and C – purple varieties.

Growth and yield parameters such as plant height, stem girth, biomass, number of leaves, length of leaves, width of leaves, number of spike per panicle, number of spikelets per spike, weight of panicle, length of panicle, width of panicle, number of grains per panicle, 1000 seeds weight, number of roots, weight of roots, and length of roots were gathered and analyzed using the Statistical Tool for Agricultural Research (STAR) software.

(11) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant height

Plant height of sorghum was highly affected with the different varieties. **Figure 1** presented that taller plants were observed from the purple varieties, followed by white sorghums which was similar to the red varieties.

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Stem girth

The different varieties influenced the stem girth of sorghum. As presented in **Figure 2**, bigger stems were noted in red varieties, similar to purple variety. The smaller stems were observed from the white sorghum.

Biomass

The result of the study revealed that the biomass of sorghum was highly affected by the different varieties. As shown in **Figure 3**, heavier plants were recorded from purple varieties. The lighter sorghum was observed from the white and red varieties.

Number of leaves

A highly significant difference was observed on the number of leaves of sorghum. **Figure 4** presented that the most number of leaves were counted in the red sorghums, followed by the purple varieties. The least number of leaves were recorded in the white varieties.

Length and width of leaves

The varieties significantly influenced the leaf length and width of sorghum. **Figures 5a and 5b** shows that longer and wider leaves were recorded from the white and purple varieties, while the red varieties resulted a shorter and narrower leaves.

Number of spike per panicle

Figure 6 revealed that the number of spike per panicle was not affected by the different varieties. Means ranges from 50.06-58.42.

Number of spikelets per spike

The different varieties negatively affected the number of spikelets per spike of sorghum (**Figure 7**). The means ranges from 116.7-163.26.

Weight of panicle

As presented in **Figure 8**, the different varieties significantly influenced the weight of panicle. Heavier panicle was exhibited by a purple sorghum, while the white and red varieties gave a lighter panicle.

Length of panicle

A highly significant effect of the different varieties was noted in the length of panicle. Longer panicle was recorded from the purple varieties which were found comparable to the red varieties, while the shorter panicle was observed in white varieties (**Figure 9**).

Width of panicle

The analysis of variance revealed a highly significant effect of the different varieties on the width of panicle. **Figure 9** shows that white and purple sorghum obtained a wider panicle, while narrower in the red varieties.

Number of grains per panicle

Grains per panicle of sorghum were not affected with the different varieties. Means in **Figure 10** ranges from 858.48-1,263.06.

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1000 seeds weight

The different varieties affected the 1000 seeds weight of sorghum (**Figure 11**). Heavier weight of 1000 seeds was recorded from the purple varieties which was similar to white varieties, however, the white varieties was also found comparable to the red sorghums that resulted a lighter 1000 seeds weight.

Number of roots

The different varieties did not showed a significant effect on the number of roots of sorghum as indicated in **Figure 12**. Means ranges from 19.98-23.78.

Weight of roots

Different varieties resulted a significant effect on the weight of roots (**Figure13**). Purple varieties exhibited a heavier root than white and red varieties.

Length of roots

Length of roots was not affected with the different varieties. The means ranges from 21.42-29.02 (**Figure 14**).

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the results of the study, the following conclusions were consolidated:

1. The different varieties significantly varies except for the number of spike per panicle, number of spikelets per spike, number of grains per panicle, number of roots, and length of roots.
2. Purple varieties exhibited tallest plants, heaviest biomass, longest and widest leaves, heaviest 1000 seeds weight, and heaviest roots, widest panicle, longest panicle and heaviest panicles among all the varieties.
3. Biggest stems and most number of leaves produced were observed from the red sorghum.
4. Among the sorghum variety, purple sorghum exhibited a remarkable result and are considered adaptable under the Capiz condition.
- 5.

Recommendations

Based on the result of the study, the following recommendations are forwarded:

1. Grow purple sorghum variety for taller plant, heaviest biomass, longest and widest leaves, heaviest 1000 seeds weight.
2. Grow red sorghum for animal feeds considering its leaf and stem size.
3. Conduct similar studies considering the following:
 - a. Use NSIC varieties
 - b. Apply recommended rates of fertilize
 - c. Lengthen observation period
 - d. Test digestibility of the different sorghum varieties

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

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(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

References

- BAR. (2017). Bureau of Agricultural Research: Sorghum Varietal Trials in the Philippines.
- Blum, A. (2004). Sorghum physiology. In Nguyen, H. & Blum, A. (Eds.), Physiology and Biotechnology Integration for Plant Breeding (pp. 141-223). CRC Press.
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- Rooney, W. L. (2000). Genetics and cytogenetics. In Smith, C. W. & Frederiksen, R. A. (Eds.), Sorghum: Origin, History, Technology, and Production (pp. 261-307). John Wiley & Sons.
- Smith, C. W., & Frederiksen, R. A. (2000). Sorghum: Origin, History, Technology, and Production. John Wiley & Sons.
- Tuinstra, M. R., Grote, E. M., Goldsbrough, P. B., & Ejeta, G. (1997). Genetic analysis of post-flowering drought tolerance and components of grain development in sorghum. Molecular Breeding, 3, 439-448.

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APPENDICES

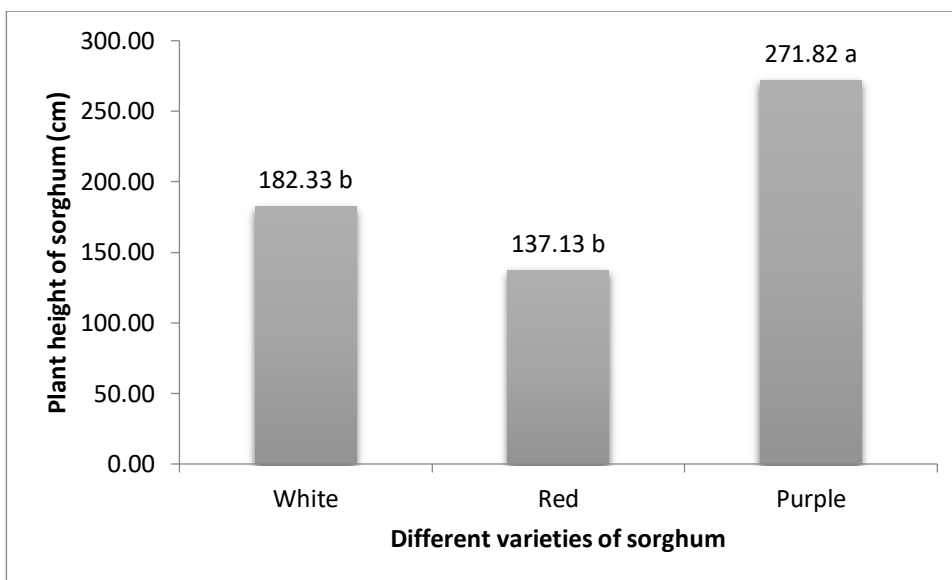


Fig. 1. Plant height (cm) of sorghum as affected with the different varieties.

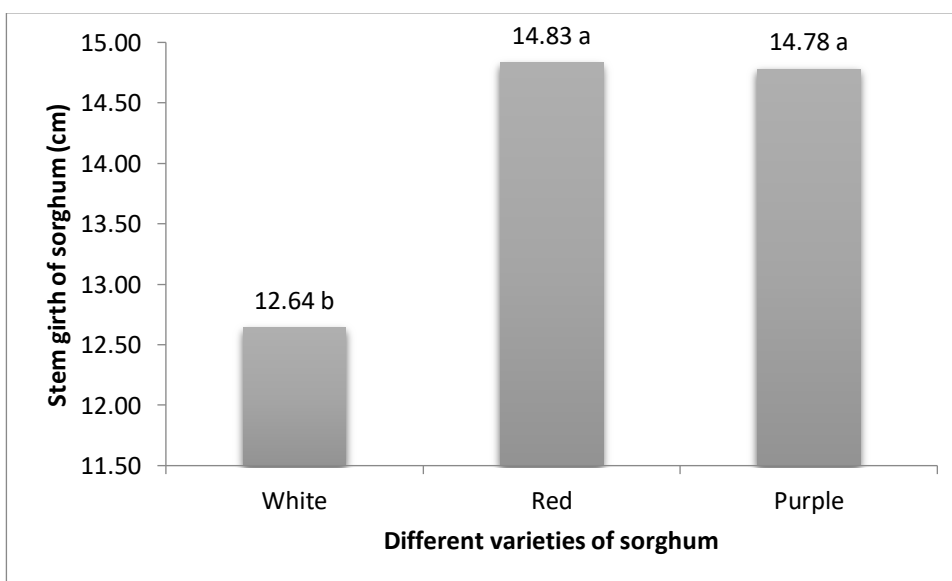


Fig. 2. Stem girth (cm) of sorghum as affected with the different varieties.

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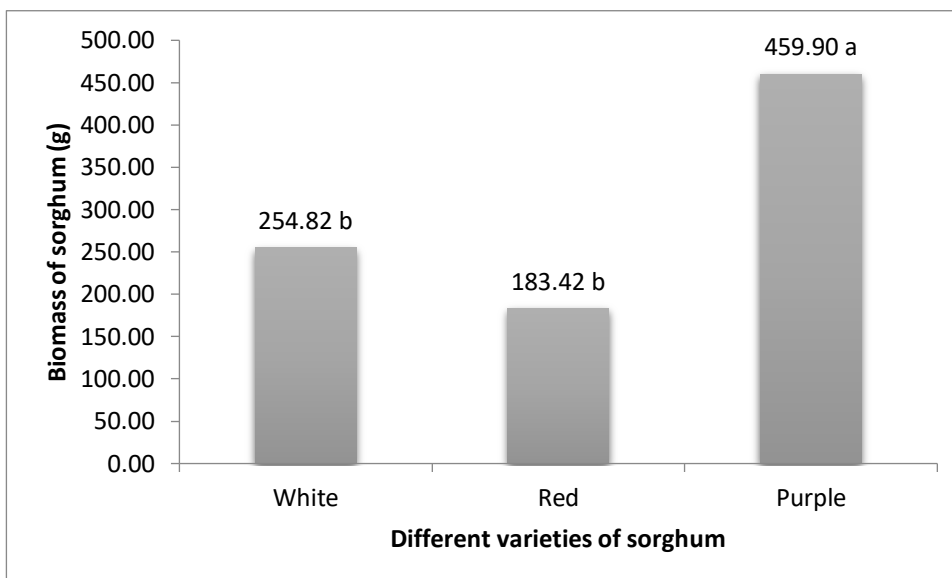


Fig. 3. Biomass (g) of sorghum as affected with the different varieties.

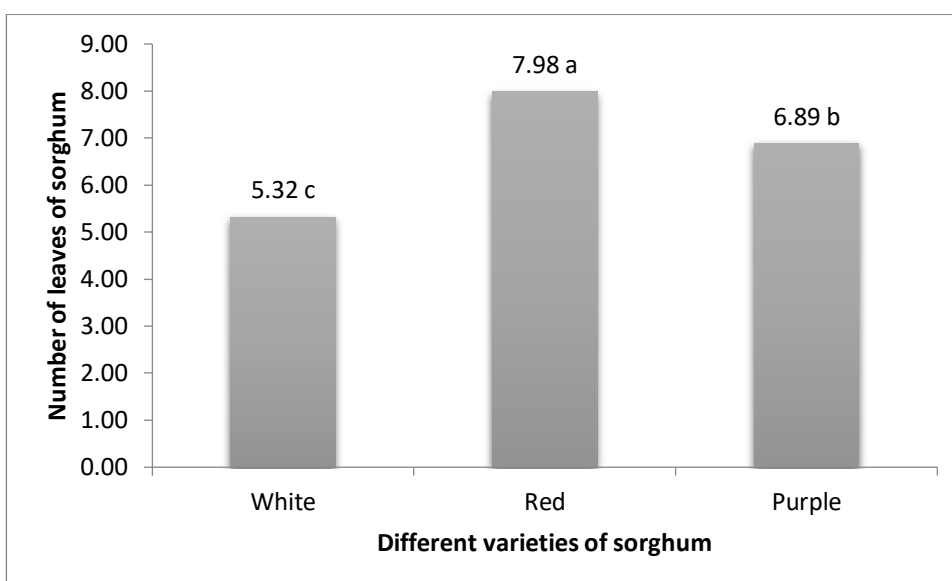


Fig. 4. Number of leaves of sorghum as affected with the different varieties.

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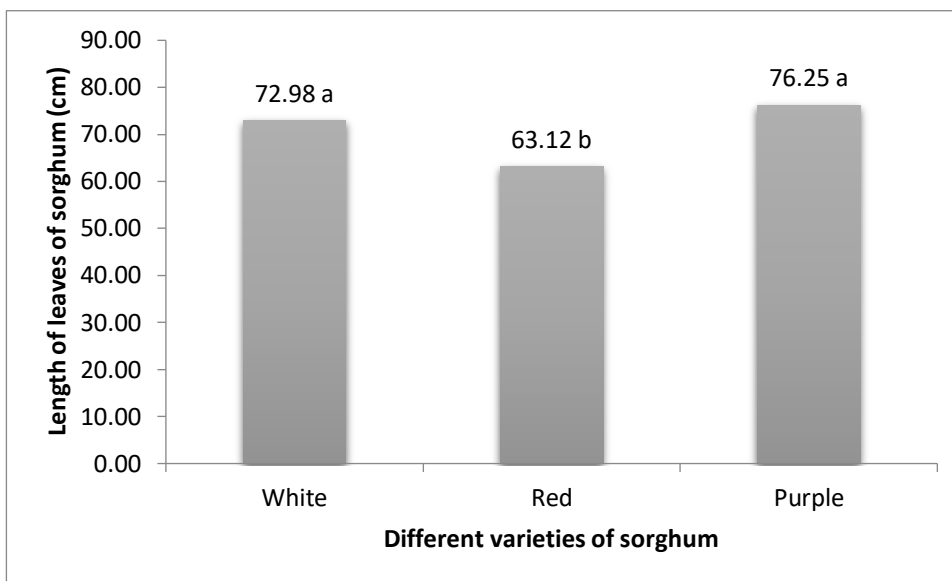


Fig. 5.a. Length (cm) of leaves of sorghum as affected with the different varieties.

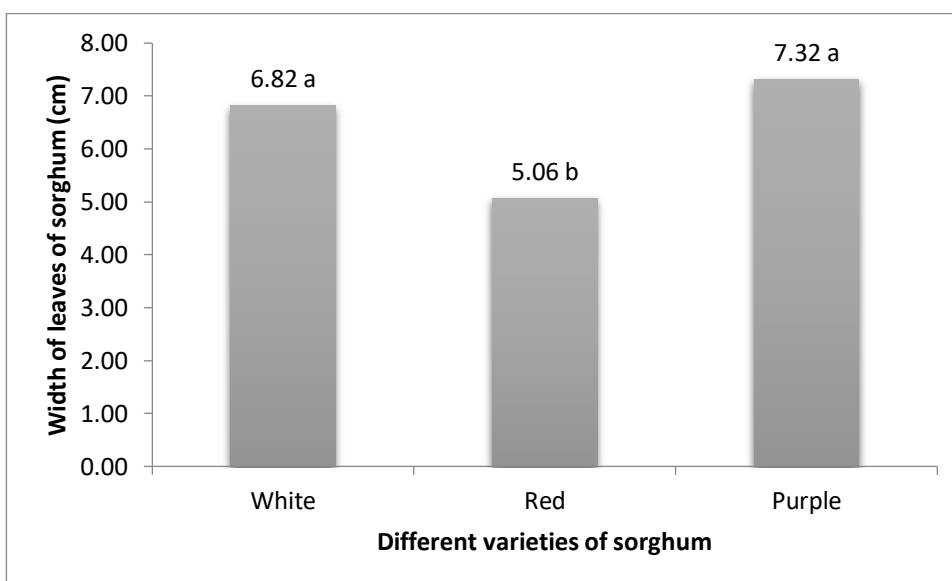


Fig. 5.b. Width (cm) of leaves of sorghum as affected with the different varieties.

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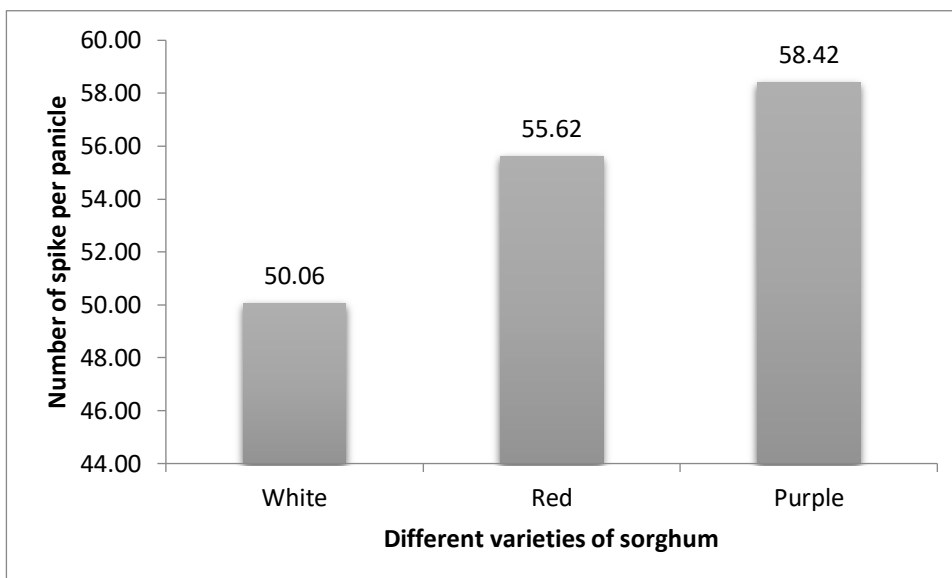


Fig. 6. Number of spike per panicle of sorghum as affected with the different varieties.

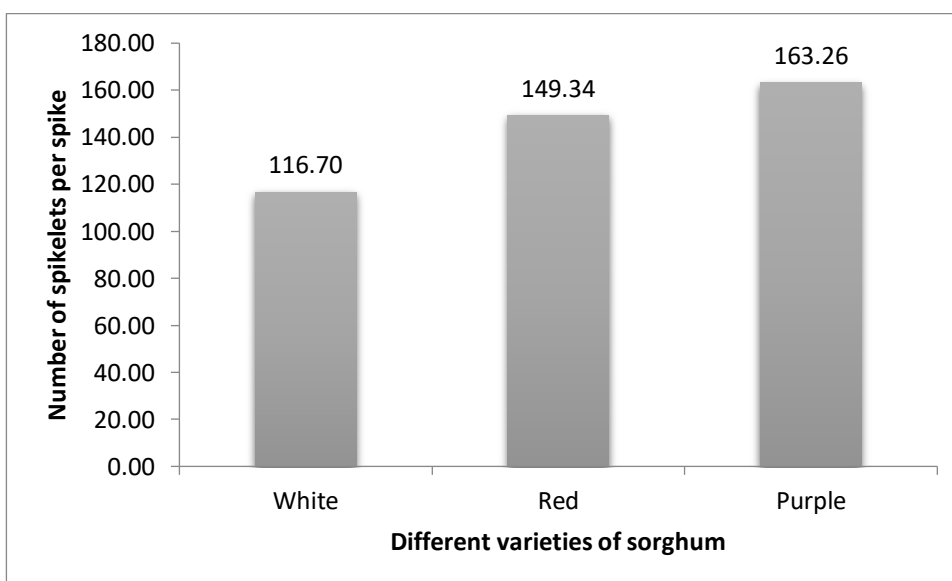


Fig. 7. Number of spikelets per spike of sorghum as affected with the different varieties.

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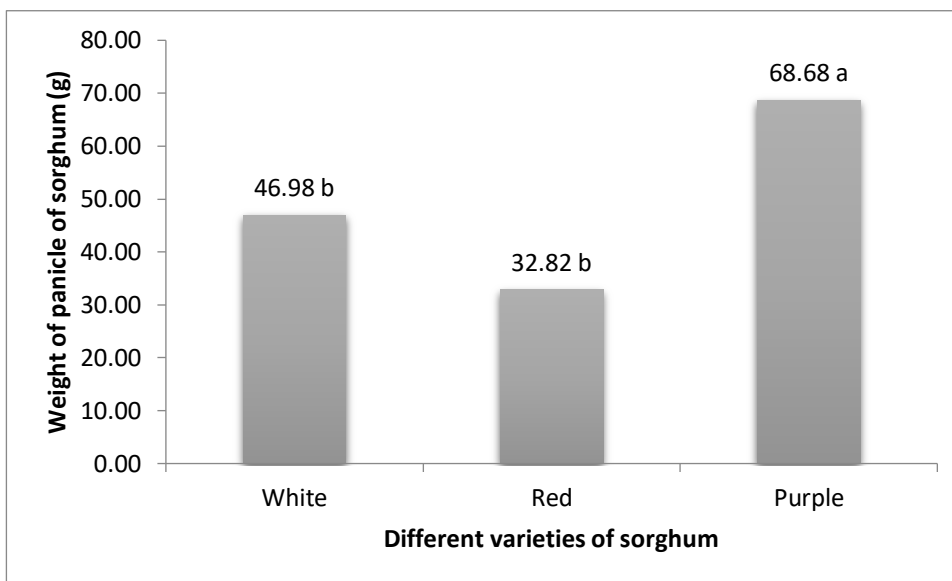


Fig. 8. Weight (g) of panicle of sorghum as affected with the different varieties.

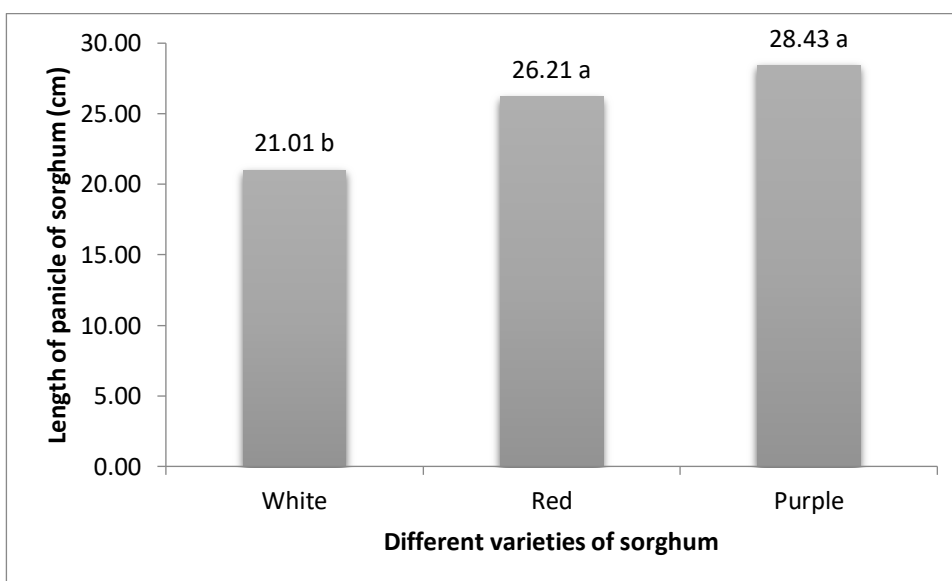


Fig. 8. Length (cm) of panicle of sorghum as affected with the different varieties.

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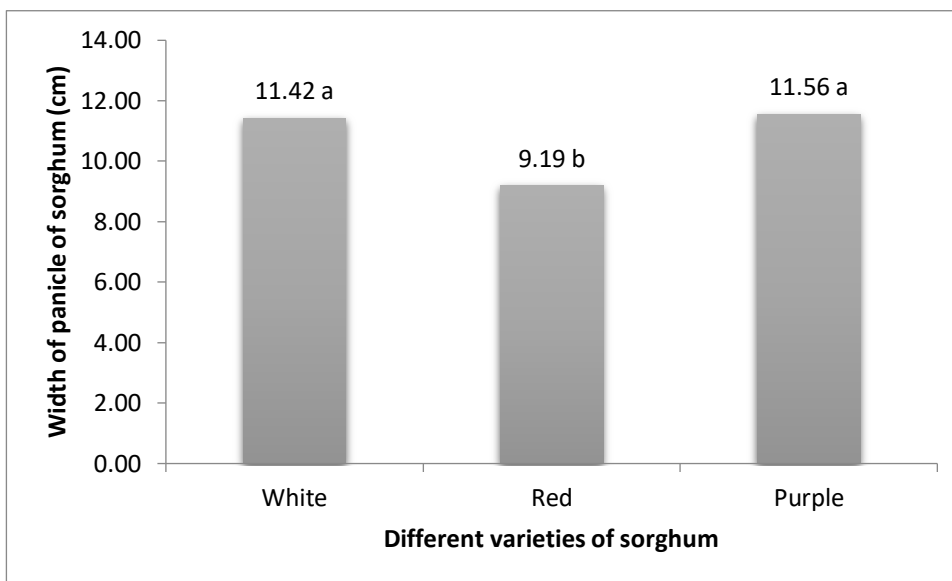


Fig. 9. Width (cm) of panicle of sorghum as affected with the different varieties.

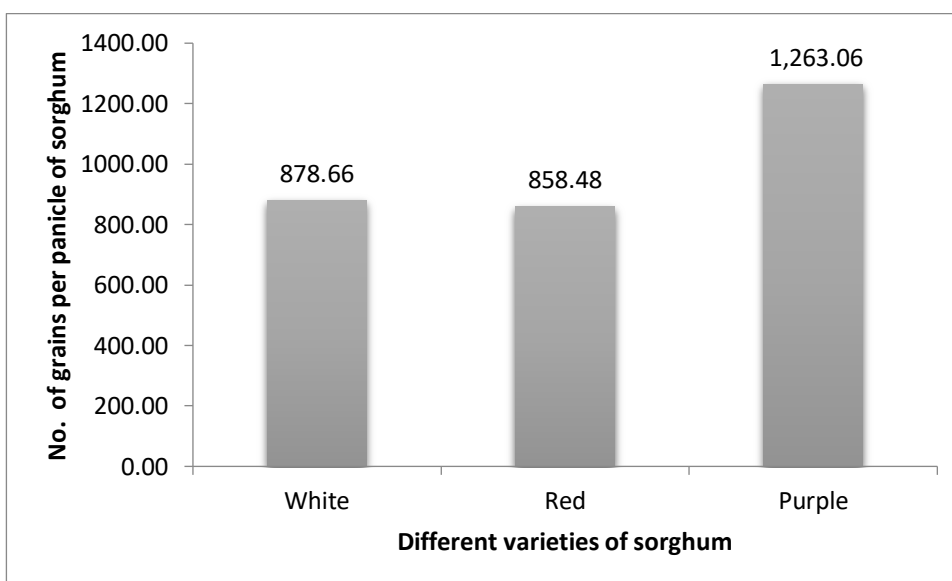


Fig. 10. Number of grains per panicle of sorghum as affected with the different varieties.

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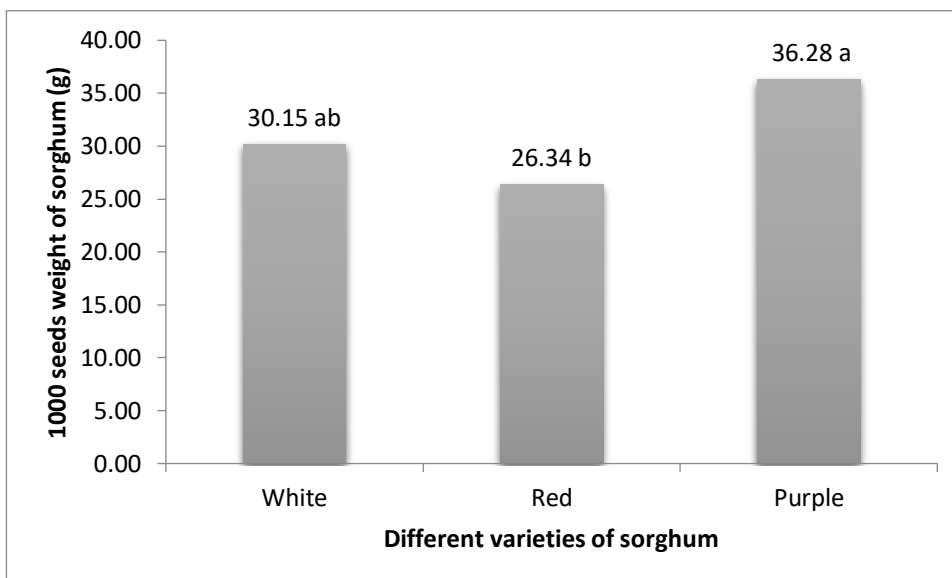


Fig. 11. Weight (g) of 1000 seeds of sorghum as affected with the different varieties.

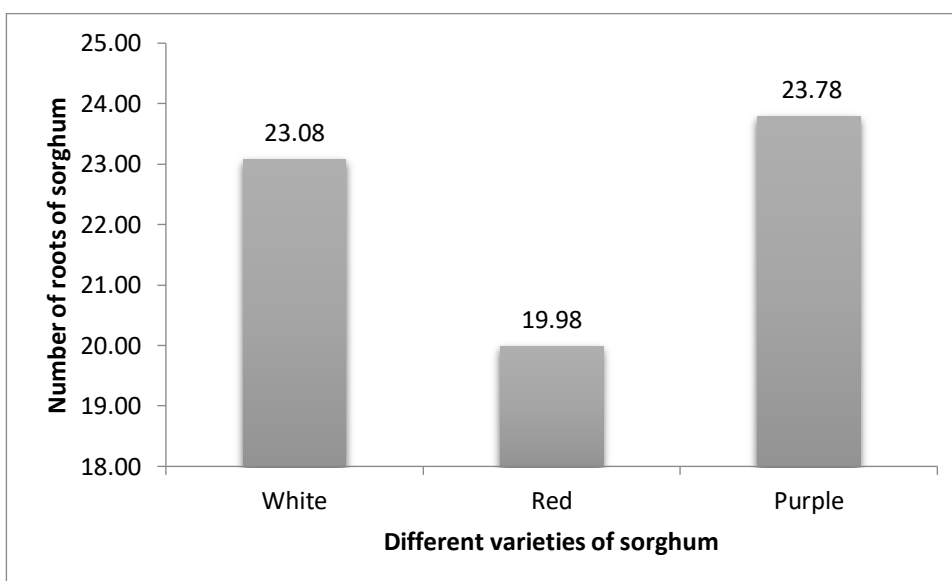


Fig. 12. Number of roots of sorghum as affected with the different varieties.

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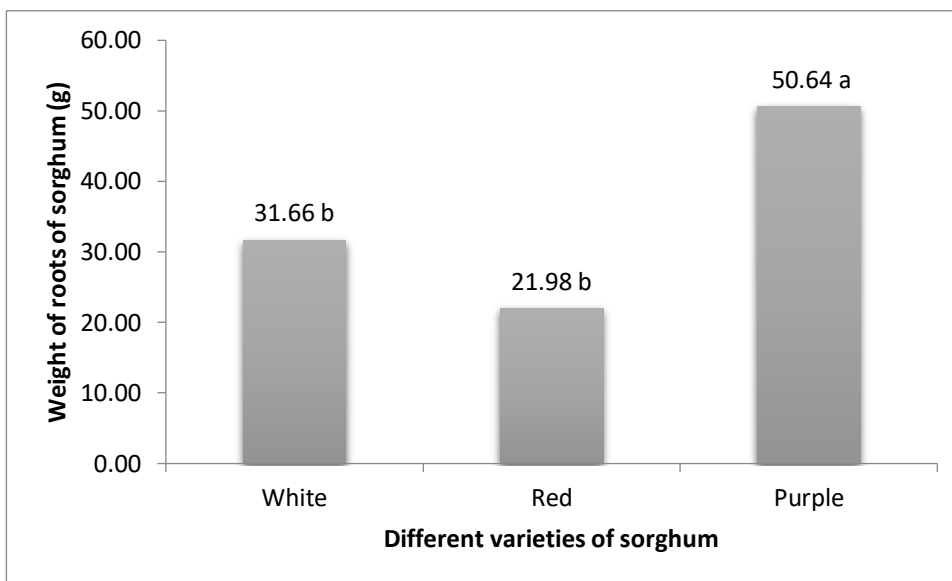


Fig. 13. Weight (g) of roots of sorghum as affected with the different varieties.

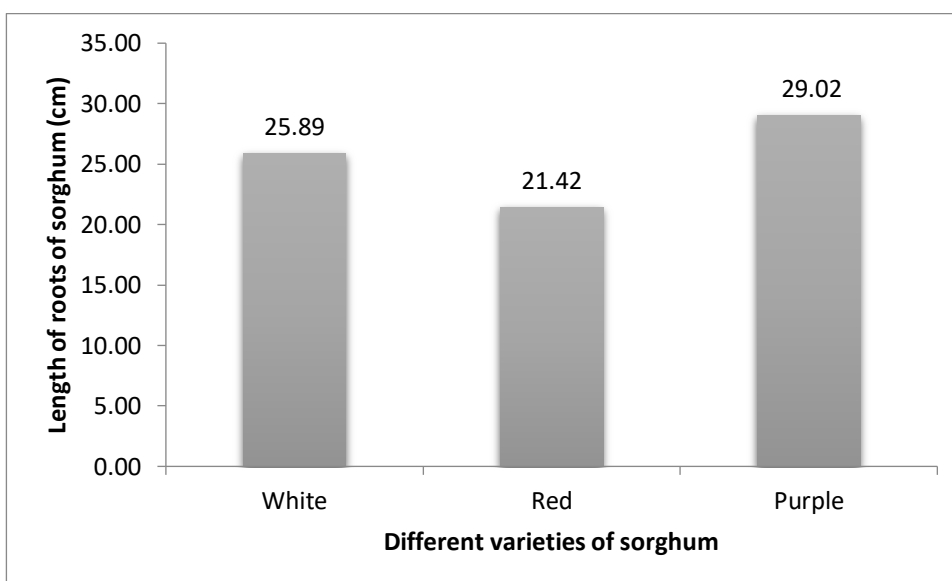


Fig. 14. Length of roots of sorghum as affected with the different varieties.